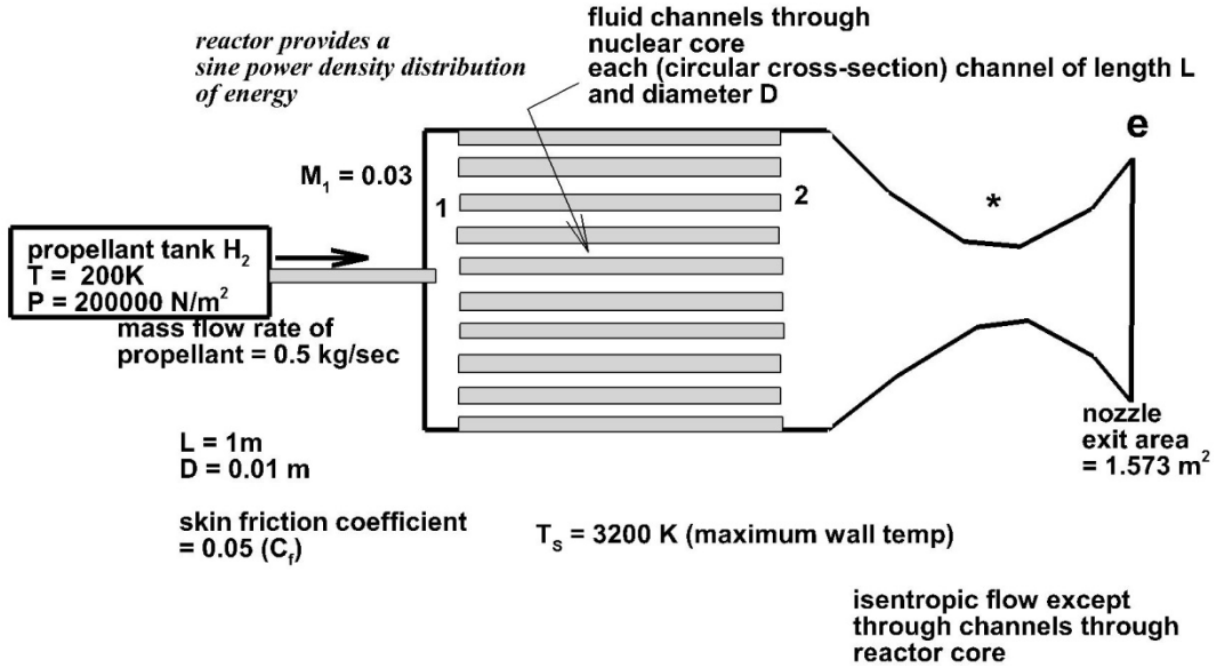
Homework 6: Nuclear Propulsion

Riley J. Kyle



Answers:

$x_{T_w=T_s} = 0.9031078m$

$\dot{Q} = 21.147MW$

$F =$

$I_{sp} =$

Conditions for sin power density distribution.

1) zero slope of T_t at $x = 0$
2) max slope of T_t at $x = L/2$
3) zero slope of T_t at $x = L$

Additionally $T_s \rightarrow max(T_w)$ occurs at some x between $x = L/2$ and $x = L$.

Can find the location where T_w is equal to T_s by using:

$$\left(\frac{x}{L}\right)_{T_w=T_s} = 1 + \frac{1}{\pi}\tan^{-1}\frac{-\pi D}{2C_f L} = 0.9031078 \tag{1}$$

Use this equation to find ΔT_m:

$$T_s = T_{t_1} + \frac{2C_F L}{\pi D}\Delta T_m\left(1 + \sqrt{1 + \left(\frac{\pi D}{2C_f L}\right)^2}\right) \rightarrow \Delta T_m = 460.152K \tag{2}$$

We can then use this equation to find the total temperature at station 2:

$$T_{t_2} - T_{t_1} = \frac{4C_F L}{\pi D}\Delta T_m \rightarrow T_{t_2} = 3129.420K \tag{3}$$

Since we know the change in temperature through the reactor we can calculate the heat generated using:

$$\dot{Q} = \dot{m}c_p\left(T_{t_2} - T_{t_1}\right) = \boxed{21.147MW} \tag{4}$$

Where:

$$c_p = \frac{\gamma}{\gamma - 1}R = 14437.5\frac{J}{kgK} \tag{5}$$